

SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI – 602105
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DATA CLEANING CHALLENGE

Course Code:	DSA05	Course Title:	Query Processing for Data Science With Real Time Applications
CO Assessed:	CO3	Assessment:	Tool 2 – Data Cleaning Challenge
Weightage:		Total Marks:	25 Marks
CO3 Statement: Identify and clean inconsistencies in data by handling missing values, duplicates, outliers, formatting issues, and applying techniques like fuzzy matching and regex. (BL5 and BL6).			
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Section / Batch:	Slot A - 2024	Date of Submission:	12-08-2026
Faculty In-charge:	K. Veena Devi	Project Mode:	Individual Submission

Code of Conduct

I _P.Sunayana(192424303)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: _P.Sunayana_

Anomaly Detection (5 Marks)	Cleaning Methods (5 Marks)	Data Transformation (5 Marks)	Analytical Comparison (5 Marks)	Governance Discussion (5 Marks)	Total (25 Marks)

Question 1: E-Commerce Customer Transactions Dataset

An online retail company has collected customer transaction data from multiple sales channels. The management wants to identify the top-performing customer segments and products. However, the dataset contains several quality issues.

CustomerID	CustomerName	Gender	Age	ProductCategory	Quantity	UnitPrice	PurchaseDate	City
C101	Raj Kumar	Male	28	Electronics	2	15000 INR	10-01-2025	Chennai
C102	Priya S	Female	35	Fashion	3	1200 INR	11-01-2025	chennai
C103	Arun V	Male	150	Electronics	1	300 USD	12-01-2025	CHENNAI
C104	Kavya R	Female	0	Grocery	-2	800 INR	15-01-2025	Coimbatore
C105	Rahul P	Male	42	Fashion	5	2500 INR	18-01-2025	Madurai
C105	Rahul P	Male	42	Fashion	5	2500 INR	18-01-2025	Madurai
NULL	Sneha M	Female	30	Electronics	2	450 USD	05-02-2025	Trichy
C107	Deepak N	Male		Grocery	3	600 INR	07-02-2025	Tiruchirappalli
C108	Meena K	Female	27	Fashion	1	1800 INR	2025-13-01	Salem
C109	John D	Male	31	Electronics	4	250 EUR	15-02-2025	Chennai

Data Issues

- Missing Customer IDs
- Duplicate transactions
- Negative quantities
- Unit prices recorded in different currencies
- Age values such as 0, 150, and blanks
- Inconsistent city names (e.g., Chennai, chennai, CHENNAI)
- Incorrect purchase dates

Tasks

1. Identify and classify all data quality issues. (5 Marks)
2. Design a cleaning strategy for each issue. (5 Marks)
3. Perform data standardization and transformation. (5 Marks)
4. Analyze how cleaning affects total revenue calculations. (5 Marks)
5. Generate a report discussing business risks if the data were not cleaned. (5 Marks)

Identify and classify all data quality issues

The dataset contains several types of data-quality problems.

Issue	Example	Classification	Impact
Missing Customer ID	NULL for Sneha M	Completeness	Transaction cannot be reliably linked to a customer
Duplicate transaction	C105 appears twice	Uniqueness	Revenue is counted twice
Negative quantity	C104 = -2	Validity	Produces negative revenue
Mixed currencies	INR, USD, EUR	Consistency	Revenue cannot be directly compared
Invalid age	0, 150	Validity	Incorrect customer segmentation

Missing age	Deepak's age blank	Completeness	Age analysis becomes unreliable
Inconsistent city names	Chennai, chennai, CHENNAI	Consistency	Same city treated as different cities
Invalid date	2025-13-01	Validity	Month 13 does not exist
Multiple date formats	10-01-2025, 2025-13-01	Consistency	Difficult to perform chronological analysis
Currency formatting	15000 INR, 300 USD	Standardization	Numerical calculations become difficult

Specific anomalies

- C105 occurs twice with exactly the same transaction information.
- C104 has quantity -2.
- C103 has age 150, which is unrealistic.
- C104 has age 0, which is invalid for a normal customer record.
- Sneha M has a missing customer ID.
- Deepak N has a missing age.
- C108 has an invalid date because month 13 does not exist.
- Chennai appears as Chennai, chennai, and CHENNAI.
- Prices use three different currencies.

2. Cleaning strategy for each issue

A systematic cleaning strategy should be applied.

Missing Customer ID

For the record:

NULL | Sneha M | Female | 30 | Electronics | 2 | 450 USD

First check whether the transaction can be linked using another reliable identifier.

If no reliable identifier exists:

- Generate a temporary transaction ID.
- Do not invent a customer ID without evidence.
- Mark the record for review.

Duplicate transactions

C105 occurs twice:

C105 | Rahul P | Male | 42 | Fashion | 5 | 2500 INR

If all fields are identical, retain only one record.

Negative quantity

C104 has:

Quantity = -2

A negative quantity is normally invalid for a sales transaction.

Possible actions:

- Check whether it represents a return/refund.
- If it is a data-entry error, correct it using the original source.
- If it represents a genuine return, store it using a separate transaction type rather than treating it as a normal sale.

Currency

Convert all currencies to one reporting currency, for example INR.

Use an approved exchange-rate table based on the transaction date.

Age

Use a validation rule such as:

$0 < \text{Age} \leq 120$

Therefore:

- Age 150 → invalid.
- Age 0 → invalid.
- Missing age → retain as NULL or impute only when justified.

City

Convert values to a standard representation:

Chennai

Chennai

Chennai

For example:

`city = city.strip().title()`

A controlled mapping table is preferable for production systems.

Dates

Convert all dates into ISO format:

YYYY-MM-DD

For example:

10-01-2025 → 2025-01-10

The invalid:

2025-13-01

should be flagged and corrected from the original source rather than automatically changing it to an assumed date.

3. Data standardization and transformation

A standardized dataset could have these fields:

CustomerID

CustomerName

Gender

Age

ProductCategory

Quantity

UnitPrice

Currency

PurchaseDate

City

Standardization rules

Gender

Male → Male

Female → Female

City

Chennai

Chennai

Chennai

Coimbatore

Madurai

Trichy

Tiruchirappalli

Salem

Trichy and Tiruchirappalli should be mapped to one official representation if they refer to the same city.

Date

Convert all dates to:

YYYY-MM-DD

Currency

Convert:

USD → INR

EUR → INR

INR → INR

using a documented exchange-rate table.

Revenue calculation

For each transaction:

$\text{Revenue} = \text{Quantity} \times \text{UnitPrice}$

For example, C101:

$2 \times ₹15,000 = ₹30,000$

C102:

$3 \times ₹1,200 = ₹3,600$

C105:

$5 \times ₹2,500 = ₹12,500$

After currency conversion, all transactions can be aggregated correctly.

4. Effect of cleaning on total revenue

The original dataset contains 10 records, but C105 is duplicated.

If the duplicate is counted, its revenue is:

$5 \times ₹2500 = ₹12,500$

Therefore, the duplicate causes revenue to be overstated by **₹12,500**.

The negative quantity for C104 gives:

$-2 \times ₹800 = -₹1,600$

This may incorrectly reduce total revenue if the transaction is actually a data-entry error.

The USD and EUR transactions cannot be added directly to INR transactions.

For example:

₹30,000 + \$300

is not a meaningful total until \$300 is converted to INR.

Therefore, cleaning can significantly change the calculated revenue.

Effect of each issue

Issue	Revenue effect
Duplicate C105	Overstates revenue
Negative quantity	May understate revenue
Mixed currencies	Makes total invalid
Missing transaction ID	Makes transaction tracking difficult
Invalid dates	Can place revenue in the wrong reporting period

Conclusion

The cleaned revenue is more reliable because:

1. Duplicates are removed.
2. Invalid quantities are investigated.
3. Currency is standardized.
4. Dates are standardized.
5. Invalid records are corrected or excluded according to documented rules.

5. Business risks if data is not cleaned

Poor-quality transaction data can cause major business problems.

1. Incorrect revenue reporting

Duplicate transactions can make sales appear higher than they really are.

2. Incorrect customer segmentation

Invalid ages such as 0 and 150 can distort age-based marketing.

3. Incorrect product performance

Negative quantities and inconsistent categories can change product rankings.

4. Incorrect regional analysis

Chennai, chennai, and CHENNAI may be treated as different cities.

5. Financial reporting errors

Mixing INR, USD, and EUR can produce incorrect revenue totals.

6. Poor decision-making

Management may incorrectly identify:

- Best customers
- Best products
- Best cities
- Best sales channels

Conclusion

Data cleaning is necessary before performing customer segmentation, revenue analysis, forecasting, or strategic decision-making.

Question 2: Hospital Patient Records Dataset

PatientID	Name	Age	Gender	AdmissionDate	DischargeDate	Diagnosis	TreatmentCost	DoctorID
P1001	Ravi Kumar	45	M	10-01-2025	15-01-2025	Diabetes	25000	D101
P1002	Priya S	38	F	12-01-2025	11-01-2025	Hypertension	18000	D102
P1003	Arun V	125	M	13-01-2025	20-01-2025	diabetes	22000	D101
NULL	Kavya R	29	F	15-01-2025	17-01-2025	Diabtes	15000	D103
P1005	Rahul P	51	M	18-01-2025	25-01-2025	Asthma	NULL	D104
P1005	Rahul P	51	M	18-01-2025	25-01-2025	Asthma	NULL	D104
P1006	Sneha M	34	F	15-01-2025	18-01-2025	Hypertension	12000	D102
P1007	Deepak N		M	01-02-2025	05-02-2025	Asthma	10000	D104
P1008	Meena K	26	F	10-02-2025	15-02-2025	HYPERTENSION	13500	D102
P1009	John D	40	M	12-02-2025	18-02-2025	Diabetes	20000	D101

A hospital wants to analyze patient admissions and treatment effectiveness. Data were entered manually by different departments, leading to inconsistencies.

Data Issues

- Missing patient IDs
- Duplicate patient records
- Discharge dates before admission dates
- Null treatment costs
- Inconsistent diagnosis names
- Multiple formats of dates
- Unrealistic patient ages (>120)

Tasks

1. Detect all inconsistencies and anomalies. (5 Marks)
2. Develop rules for handling missing and invalid values. (5 Marks)
3. Create a process to standardize diagnosis names. (5 Marks)
4. Evaluate how cleaning changes average treatment cost and hospital stay duration. (5 Marks)
5. Recommend data governance measures to prevent future errors. (5 Marks)

Hospital Patient Records Dataset

1. Detect all inconsistencies and anomalies

Issue	Example	Type
Missing Patient ID	Kavya R = NULL	Completeness
Duplicate record	P1005 appears twice	Uniqueness
Invalid admission/discharge sequence	P1002: discharge 11-01, admission 12-01	Validity
Missing treatment cost	P1005	Completeness
Inconsistent diagnosis	Diabetes, diabetes, Diabtes	Consistency
Different date formats	10-01-2025, etc.	Standardization
Unrealistic age	Arun = 125	Validity
Missing/incorrect name formatting	Names split across lines	Formatting
Diagnosis capitalization	HYPERTENSION	Consistency

Important anomalies

P1002 has:

Admission = 12-01-2025

Discharge = 11-01-2025

This is impossible because the discharge date precedes admission.

Arun V has age:

125

which is outside the reasonable human age range.

Diabetes, diabetes, and Diabtes represent inconsistent diagnosis values.

2. Rules for missing and invalid values

Missing Patient ID

Do not randomly assign an ID.

Use:

1. Hospital registration system.
2. Source record.
3. Patient master database.

If the ID cannot be recovered, assign a temporary review status.

Duplicate records

P1005 occurs twice with identical information.

Keep one record after verifying that it is a true duplicate.

Invalid age

Define:

$0 < \text{Age} \leq 120$

Age 125 should be flagged.

Missing treatment cost

Possible approaches:

- Retrieve from billing records.
- If unavailable, leave as NULL.
- Impute only when statistically justified.

For financial reporting, fabricated values should not be inserted without evidence.

Invalid dates

Require:

DischargeDate ≥ AdmissionDate

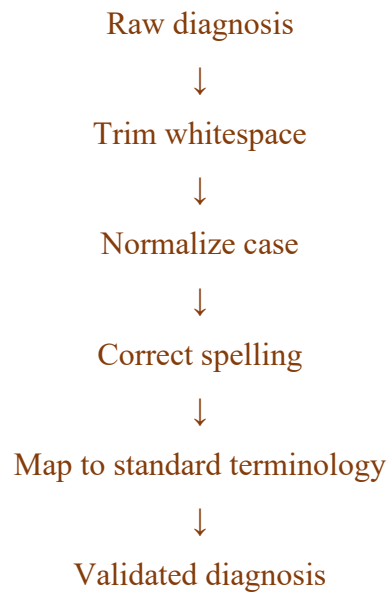
P1002 should therefore be flagged for correction.

3. Standardizing diagnosis names

A diagnosis mapping table can be created:

Raw value	Standard value
Diabetes	Diabetes
diabetes	Diabetes
Diabtes	Diabetes
Hypertension	Hypertension
HYPERTENSION	Hypertension
Asthma	Asthma

The process is:



For example:

"diabetes" → "Diabetes"

"Diabtes" → "Diabetes"

"HYPERTENSION" → "Hypertension"

For a real healthcare system, diagnosis names should preferably be mapped to an authoritative coding system rather than relying only on spelling corrections.

4. Effect of cleaning on average treatment cost and hospital stay

Treatment cost

The duplicate P1005 has NULL treatment cost, so it does not directly increase the total cost.

Known costs are:

$$25000+18000+22000+15000+12000+10000+13500+20000 = ₹135,500$$

There are 8 records with known treatment costs.

Therefore:

$$\text{Average} = 135500 / 8 = ₹16,937.50$$

The missing P1005 cost should not be automatically treated as zero because zero would incorrectly reduce the average.

Hospital stay

For valid records:

$$\text{StayDuration} = \text{DischargeDate} - \text{AdmissionDate}$$

For example:

P1001:

10-Jan → 15-Jan

Duration = 5 days.

P1002 has:

12-Jan → 11-Jan

which produces an invalid negative duration.

Therefore, it must be corrected or excluded before calculating the average.

Effect of cleaning

Before cleaning:

- Invalid negative stay duration may distort the average.
- Duplicate records may distort patient-level statistics.
- Missing costs may distort averages if treated incorrectly.

After cleaning:

- Invalid dates are removed/corrected.
- Duplicate records are removed.

- Missing costs are handled transparently.

Therefore, the cleaned average treatment cost and stay duration provide a more reliable measure of hospital performance.

5. Data governance measures

The hospital should introduce:

1. Data validation rules

Examples:

Age: 0–120

Treatment cost: ≥ 0

Discharge $\geq$ Admission

Patient ID: mandatory and unique

2. Controlled vocabularies

Diagnosis names should come from a predefined list or coding system.

3. Mandatory fields

Patient ID, admission date, and essential clinical information should not be allowed to remain blank.

4. Duplicate detection

The system should automatically detect repeated patient IDs and identical records.

5. Standard date format

Store dates internally using ISO format:

YYYY-MM-DD

6. Audit trails

Every correction should record:

- Original value
- Corrected value
- User
- Date/time
- Reason

7. Staff training

Employees entering patient data should be trained in data-quality standards.

Question 3: Banking Loan Approval Dataset

A bank uses customer data to determine loan eligibility. The dataset contains errors that may affect risk assessment and approval decisions.

Customer ID	AnnualIncome	CreditScore	LoanAmount	EmploymentStatus	Age	ExistingDebt	LoanStatus
L001	750000	720	300000	Full Time	35	50000	Approved
L002	900K	680	400000	FT	42	100000	Approved
L003	1200000	NULL	600000	fulltime	28	200000	Approved
L004	150000	450	-200000	Part Time	30	50000	Rejected
L005	5000000	790	1000000	Self Employed	16	10000	Approved
L006	450000	350	250000	Unemployed	40	300000	Rejected
L007	75000000	800	500000	Full Time	45	100000	Approved
L008	650000	690	350000	FT	33	70000	Approved
L008	650000	690	350000	FT	33	70000	Approved
L010	NULL	710	300000	Full Time	37	80000	Approved

Data Issues

- Missing credit scores

- Income values recorded in different units
- Negative loan amounts
- Duplicate customer IDs
- Age values below 18
- Employment status entered differently (FT, Full Time, fulltime)
- Extreme outliers in income

Tasks

1. Identify critical issues that could impact loan decisions. (5 Marks)
2. Propose cleaning techniques for missing and inconsistent data. (5 Marks)
3. Detect and justify treatment of outliers. (5 Marks)
4. Analyze the effect of cleaning on loan approval rates. (5 Marks)
5. Discuss ethical implications of incorrect data cleaning decisions. (5 Marks)

Banking Loan Approval Dataset

1. Critical issues affecting loan decisions

The dataset contains errors that can directly affect credit risk assessment.

Missing credit score

L003 has:

CreditScore = NULL

Credit score is an important loan decision variable.

Mixed income units

Examples:

750000

900K

900K means approximately 900,000, but it must be standardized.

Negative loan amount

L004:

LoanAmount = -200000

This is invalid.

Duplicate customer

L008 occurs twice.

This can cause the customer to be counted twice.

Age below 18

L005:

Age = 16

This may violate lending eligibility rules.

Employment inconsistencies

Full Time

FT

fulltime

represent the same employment status but are stored differently.

Extreme income

L007:

75,000,000

is much higher than most other records and requires investigation.

2. Cleaning techniques

Credit score

If the score is missing:

- Retrieve it from the source system.
- If unavailable, keep it missing and send the application for review.
- Avoid automatically assigning an average credit score for an individual loan decision.

Income

Convert all values into a single numerical unit.

For example:

900K → 900000

Loan amount

Apply:

LoanAmount > 0

L004 should be flagged.

Employment status

Create a mapping:

Duplicate customers

L008 appears twice.

Raw value	Standard
FT	Full Time
Full Time	Full Time
fulltime	Full Time
Part Time	Part Time
Self Employed	Self Employed

Verify whether the records represent the same application or repeated data before removing one.

Age

Set a business validation rule:

Age ≥ 18

L005 should therefore be reviewed.

3. Detecting and treating outliers

The income values include:

150000

450000

500000

650000

750000

900000

1200000

5000000

75000000

The 75,000,000 value is an extreme outlier compared with most observations.

However, an outlier is not automatically an error.

Possible reasons

It could represent:

- A genuine high-income customer.
- A unit error.

- A data-entry error.
- A corporate/business customer incorrectly entered into an individual-customer dataset.

Recommended treatment

First investigate the original source.

If confirmed as an error:

- Correct it from the source.

If it is genuine:

- Retain it.
- Use robust statistical methods.

Possible techniques include:

- IQR method
- Median and MAD
- Winsorization for analytical models, when appropriate

For loan approval, simply deleting a high-income customer could create unfair decisions.

4. Effect of cleaning on loan approval rate

The original data contains:

Approved = 7

Rejected = 2

There are 9 non-null loan-status records.

Therefore:

Approval Rate = $97 \times 100 = 77.78\%$

However, this rate is affected by data-quality problems.

For example:

- L003 has a missing credit score.
- L004 has a negative loan amount.
- L005 is underage.
- L008 is duplicated.
- L007 has an extreme income value.

If invalid records are removed, the denominator changes.

Therefore, the cleaned approval rate may differ significantly from 77.78%.

Important point

The exact post-cleaning rate depends on the organization's rules.

For example:

- If L005 is excluded due to age, the rate changes.
- If L008's duplicate is removed, the count changes.
- If L003 remains after retrieving its credit score, it remains in the analysis.

Therefore, cleaning decisions must be documented before calculating approval statistics.

5. Ethical implications

Data cleaning in banking has serious ethical implications because it can affect whether people receive loans.

Risk of unfair rejection

If missing information is automatically interpreted as poor credit, legitimate customers may be rejected.

Risk of unfair approval

If invalid or inflated income is retained, a customer may appear financially stronger than they really are.

Bias

Incorrect cleaning may disproportionately affect certain customer groups.

Transparency

Banks should document:

- Why a value was changed.
- Which values were removed.
- Which values were imputed.
- How decisions were affected.

Human review

Borderline cases should be reviewed rather than automatically rejected based on questionable data.

Conclusion

Data cleaning must preserve accuracy and fairness. The objective should not be to manipulate the dataset to achieve a desired approval rate.

Question 4: Smart City Traffic Monitoring Dataset

A smart city project collects traffic sensor data to optimize traffic signals. Sensor malfunctions and transmission errors have affected data quality.

Sensor ID	Timestamp	VehicleCount	AverageSpeed	TrafficSignalStatus	RoadName	WeatherCondition
S101	01-01-2025 08:00	120	45	Green	Anna Salai	Sunny
S102	01-01-2025 08:05	135	52	Red	anna salai	Sunny
S103	01-01-2025 08:10	-50	48	Green	ANNA SALAI	Cloudy
S104	01-01-2025 08:15	140	350	Yellow	GST Road	Rainy
S105	NULL	110	42	Green	GST ROAD	Sunny
S106	01-01-2025 08:25	130	46	Red	OMR	NULL
S106	01-01-2025 08:25	130	46	Red	OMR	NULL
S107	01-01-2025 08:30	150	55	Green	Old Mahabalipuram Road	Sunny

S10A	01-01- 2025 08:35	145	53	Green	OMR	Cloudy
S109	01-01- 2025 08:40	138	50	Green	GST Road	Rainy

Data Issues

- Missing timestamps
- Duplicate sensor readings
- Vehicle counts recorded as negative numbers
- Average speed exceeding 300 km/h
- Missing weather information
- Inconsistent road naming conventions
- Sensor IDs with formatting errors

Tasks

1. Assess the quality of the traffic dataset. (5 Marks)
2. Recommend cleaning techniques for time-series data. (5 Marks)
3. Design rules to identify faulty sensor readings. (5 Marks)
4. Evaluate how cleaning affects congestion analysis. (5 Marks)
5. Suggest automated monitoring techniques for future data collection. (5 Marks)

Smart City Traffic Monitoring Dataset

1. Assess the quality of the traffic dataset

The dataset contains several quality problems.

Problem	Example	Type
Missing timestamp	S105	Completeness
Duplicate reading	S106	Uniqueness
Negative vehicle count	S103 = -50	Validity
Impossible speed	S104 = 350 km/h	Validity
Missing weather	S106	Completeness

Inconsistent road names	Anna Salai, anna salai, ANNA SALAI	Consistency
Incorrect sensor ID	S10A	Validity
Formatting problems	Multi-line values	Standardization

Sensor ID

Most IDs follow:

S + three digits

For example:

S101

S102

S103

But:

S10A

contains a letter where a digit is expected.

2. Cleaning techniques for time-series data

Traffic data is time-dependent, so the cleaning process should preserve chronological order.

Step 1 – Standardize timestamps

Convert:

01-01-2025 08:00

to:

2025-01-01 08:00:00

Step 2 – Sort by timestamp

Sort all readings chronologically.

Step 3 – Detect duplicate observations

S106 appears twice with the same timestamp and values.

Keep one after verification.

Step 4 – Detect missing timestamps

S105 has a missing timestamp.

The timestamp should be recovered from the sensor system if possible.

Do not simply invent a timestamp.

Step 5 – Handle invalid values

Vehicle count cannot normally be negative.

Therefore:

-50 → invalid

Speed 350 km/h is suspicious and should be flagged.

Step 6 – Handle missing weather

The missing value can be:

- Retrieved from another sensor.
- Matched with an official weather dataset.
- Kept as NULL if unavailable.

3. Rules to identify faulty sensor readings

Vehicle count

$\text{VehicleCount} \geq 0$

Therefore:

-50 → faulty

Speed

A reasonable operational threshold can be defined based on the road type and sensor configuration.

For example:

$0 \leq \text{Speed} \leq 200 \text{ km/h}$

could be used as a broad screening rule.

Thus:

350 km/h → faulty/suspicious

The exact threshold should be based on engineering requirements.

Sensor ID

Expected format:

$^Sd_{\{3\}}\$$

Therefore:

S101 → valid

S10A → invalid

Duplicate readings

A duplicate can be identified using:

SensorID + Timestamp

If the same sensor reports exactly the same timestamp and values twice, it should be flagged.

Missing timestamp

Any record without a timestamp should be placed in a quarantine/review queue.

4. Effect of cleaning on congestion analysis

Congestion analysis commonly uses:

- Vehicle count
- Average speed
- Time
- Road
- Weather
- Signal status

Faulty values can significantly distort the results.

Negative vehicle count

A value of -50 can artificially reduce average traffic volume.

Speed of 350 km/h

This can artificially increase average speed.

For example, suppose normal speeds are:

45, 52, 48, 42, 46, 55

Adding 350 dramatically increases the average.

Therefore, the system may incorrectly conclude that traffic is moving quickly.

Duplicate S106

The duplicate can give excessive weight to that observation.

Missing timestamps

A reading without a time cannot be correctly assigned to a traffic period.

Conclusion

Cleaning produces more reliable:

- Peak-hour congestion estimates
- Road comparisons
- Traffic forecasts
- Signal optimization decisions

5. Automated monitoring techniques

A smart-city system should implement automated data-quality monitoring.

1. Range checks

Automatically detect:

VehicleCount < 0

Speed < 0

Speed > threshold

2. Schema validation

Check sensor IDs against:

$\text{^S\d{3}\$}$

3. Duplicate detection

Monitor:

SensorID + Timestamp

for duplicate readings.

4. Missing-value monitoring

Generate alerts when:

- Timestamp is missing.
- Weather is missing.
- Sensor readings stop arriving.

5. Time-series anomaly detection

Use:

- Moving averages
- Rolling standard deviation
- Z-scores
- IQR
- Change-point detection

6. Sensor heartbeat monitoring

If a sensor normally sends data every 5 minutes but stops transmitting, generate an alert.

7. Data-quality dashboard

Display:

- Missing values

- Invalid readings
- Duplicate percentage
- Sensor availability
- Number of anomalies

This allows administrators to detect sensor problems quickly.

Question 5: Global Sales Performance Dataset

A multinational company has merged sales data from five countries. Management wants to evaluate regional performance and forecast future sales. The merged dataset contains severe data quality problems.

Region	Country	SalesPerson	Product	SalesAmount	Currency	SalesDate	CustomerRating
Asia	India	Raj Kumar	Laptop	50000	INR	10-01-2025	4
Asia	India	Rajkumar	Laptop	55000	INR	10-01-2025	5
Europe	Germany	Anna M	Laptop	1200	EUR	12-01-2025	4
Europe	Germany	Anna M	Laptop	1200	EUR	12-01-2025	4
North America	USA	John D	Laptop	1500	USD	01-15-2025	6
Asia	India	Priya S	Mobile	NULL	INR	18-01-2025	3
Europe	France	Pierre L	Ordinateur Portable	1300	EUR	20-01-2025	4
Asia	India	Rahul P	Mobile	9999999	INR	25-01-2025	5

UK	UK	David T	Laptop	1100	GBP	28-01-2025	2
Asia	India	Priya S	Smartphone	35000	INR	30-01-2025	0

Data Issues

- Multiple currencies (USD, EUR, INR, GBP)
- Missing sales amounts
- Duplicate regional entries
- Customer ratings outside the scale (1–5)
- Different date formats across countries
- Product names in multiple languages
- Salespersons recorded with spelling variations
- Extreme sales outliers due to data-entry errors

Tasks

1. Develop a comprehensive data-cleaning framework for multinational datasets. (5 Marks)
2. Design methods to standardize currencies, products, and dates. (5 Marks)
3. Handle missing values and outliers using appropriate analytical techniques. (5 Marks)
4. Compare regional sales rankings before and after cleaning. (5 Marks)
5. Prepare an executive report explaining how data quality influences strategic business decisions. (5 Marks)

Global Sales Performance Dataset

1. Comprehensive data-cleaning framework

A multinational sales dataset requires a structured framework.

Step 1 – Data profiling

Identify:

- Missing values
- Duplicates
- Invalid values

- Inconsistent formats
- Outliers

Step 2 – Standardize country and region names

For example:

Germany → Germany

and ensure country-region mappings are consistent.

Step 3 – Standardize salesperson names

For example:

Raj Kumar

Rajkumar

should be investigated to determine whether they refer to the same salesperson.

A master employee table should be used.

Step 4 – Standardize products

Map multilingual product names:

Laptop → Laptop

Ordinateur Portable → Laptop

Mobile → Mobile

Smartphone → Smartphone

Step 5 – Standardize currencies

Convert all sales into a common reporting currency using documented exchange rates.

Step 6 – Standardize dates

Convert all dates into:

YYYY-MM-DD

Step 7 – Handle missing values

Investigate missing sales amounts.

Step 8 – Detect outliers

Investigate extremely large sales values.

2. Standardizing currencies, products, and dates

Currency standardization

The dataset contains:

INR

EUR

USD

GBP

Choose a reporting currency such as USD.

Then:

$\text{SalesUSD} = \text{SalesLocal} \times \text{ExchangeRate}$

Exchange rates should be based on the sales date or the organization's approved accounting rate.

Product standardization

Create a mapping table:

Raw product	Standard product
Laptop	Laptop
Ordinateur Portable	Laptop
Mobile	Mobile
Smartphone	Smartphone

This allows products to be compared across countries.

Date standardization

The dataset contains:

10-01-2025

12-01-2025

01-15-2025

The dates should be converted into an unambiguous internal format:

2025-01-10

2025-01-12

2025-01-15

Using ISO format avoids ambiguity.

3. Missing values and outliers

Missing sales amount

Priya S has:

$\text{SalesAmount} = \text{NULL}$

Possible approaches:

1. Retrieve the value from the original sales system.
2. Check the invoice/order database.
3. If unavailable, retain NULL for financial reporting.
4. For predictive analysis, use carefully documented imputation only when appropriate.

A missing sales amount should not automatically be replaced by zero because zero means an actual zero-value sale.

Customer ratings

The valid range is:

1–5

Therefore:

Rating = 6 → invalid

Rating = 0 → invalid

These values should be corrected from the source or marked missing.

Extreme sales outlier

The value:

9,999,999 INR

is extremely high relative to the other Indian sales.

It should be investigated.

Possible explanations:

- Genuine bulk sale
- Data-entry error
- Extra zeroes
- Unit conversion error

Do not automatically delete it.

If it is confirmed to be an error, correct it from the source.

If genuine, retain it but consider robust methods for statistical analysis.

4. Regional sales rankings before and after cleaning

Before cleaning

The raw dataset can produce misleading regional rankings because:

- Germany has a duplicate entry.
- India has an extreme 9,999,999 INR value.

- USA uses USD.
- Europe uses EUR.
- UK uses GBP.
- One Indian transaction has a missing amount.
- Product and salesperson names are inconsistent.

Therefore, simply adding raw sales values across regions is invalid.

After cleaning

The ranking should be calculated using:

1. Standardized currencies.
2. Removed duplicates.
3. Corrected invalid ratings.
4. Corrected or investigated outliers.
5. Standardized countries and regions.
6. Corrected missing values where authoritative values are available.

The resulting ranking may differ substantially from the raw ranking.

Example of the effect

The Indian outlier:

9,999,999 INR

can make Asia appear to be the strongest region.

If this value is an entry error and is corrected, Asia's ranking may fall.

Similarly, the duplicated German transaction artificially increases Europe's sales.

Conclusion

Regional ranking should always be based on the **cleaned, standardized dataset**, not raw sales totals.

5. Executive report: impact of data quality on strategic decisions

Executive Summary

The global sales dataset contains significant quality problems involving currencies, missing values, duplicates, ratings, dates, product names, salesperson names, and extreme values. If these issues are not corrected, management may make incorrect decisions about regional performance, products, employees, and future sales.

Key business risks

1. Incorrect regional performance

Different currencies make direct comparisons invalid.

2. Incorrect sales rankings

Duplicate transactions and extreme values can make a region appear stronger than it really is.

3. Incorrect salesperson evaluation

Raj Kumar and Rajkumar may be treated as different employees, causing incorrect performance measurement.

4. Incorrect product analysis

Ordinateur Portable and Laptop may be treated as separate products.

5. Forecasting errors

Outliers and missing values can distort forecasting models.

6. Customer satisfaction analysis

Ratings of 0 and 6 fall outside the valid 1–5 scale and can distort customer satisfaction metrics.

Recommended governance

The organization should implement:

- Centralized master-data management
- Standard currency tables
- Controlled product dictionaries
- Unique salesperson IDs
- ISO date formats
- Automated validation
- Duplicate detection
- Outlier monitoring
- Data-quality dashboards
- Audit trails
- Regular data-quality reviews

Final conclusion

Data quality directly influences strategic decisions. Clean and standardized data allows management to accurately compare regions, identify high-performing products, evaluate salespeople, forecast future sales, and allocate resources. Therefore, data cleaning should be treated as a continuous governance process rather than a one-time preprocessing activity.